Worksheet 1.1 — Arithmetic Sequences & Series

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

An arithmetic sequence has a constant common difference d between consecutive terms.

- General term: $u_n = u_1 + (n-1)d$.
- Sum of the first n terms: $S_n = \frac{n}{2}(2u_1 + (n-1)d) = \frac{n}{2}(u_1 + u_n)$.
- $d > 0$ increasing, $d < 0$ decreasing, $d = 0$ constant.

The Big Picture

Two equivalent sum formulas

$S_n = \frac{n}{2}(2u_1 + (n-1)d)$ uses only u_1 and d; $S_n = \frac{n}{2}(u_1 + u_n)$ is faster once you already know the last term u_n.

Finding u_1, d from two terms

Two given terms give two equations in u_1 and d — subtract them to eliminate u_1 and solve for d first.

Worked Examples

Example 1 — Find a term

Find u_{20} for the sequence $8, 13, 18, \ldots$

Solution. $u_1 = 8$, $d = 5$. $u_{20} = 8 + (19)(5) = 8 + 95$.

Conclusion: $u_{20} = 103$.

Example 2 — Solve for the first term and common difference

An arithmetic sequence has $u_6 = 23$ and $u_{11} = 48$. Find u_1 and d.

Solution. **Step 1:** Subtract: $u_{11} - u_6 = 5d = 48 - 23 = 25 \Rightarrow d = 5$.

Step 2: $u_6 = u_1 + 5d = 23 \Rightarrow u_1 + 25 = 23 \Rightarrow u_1 = -2$.

Conclusion: $u_1 = -2$, $d = 5$.

Example 3 — Sum of a series

Find S_{25} for $7, 11, 15, \ldots$

Solution. $u_1 = 7$, $d = 4$, $n = 25$. $S_{25} = \frac{25}{2}(2(7) + 24(4)) = \frac{25}{2}(14 + 96)$.

Conclusion: $S_{25} = \frac{25}{2}(110) = 1375$.

Example 4 — Application — stacking

Logs are stacked with 22 in the bottom row, each row above having 2 fewer, down to 4 in the top row. Find the total number of logs.

Solution. **Step 1:** $u_1 = 22$, $d = -2$, last term $u_n = 4$.

Step 2: Find n: $4 = 22 + (n-1)(-2) \Rightarrow -18 = -2(n-1) \Rightarrow n = 10$.

Step 3: $S_{10} = \dfrac{10}{2}(22 + 4) = 5(26)$.

Conclusion: 130 logs.

Example 5 — Sum from first and last term only

An arithmetic sequence has $u_1 = 5$, $d = 6$, and last term $u_n = 95$. Find S_n.

Solution. **Step 1:** Find n: $95 = 5 + 6(n-1) \Rightarrow 90 = 6(n-1) \Rightarrow n = 16$.

Step 2: $S_{16} = \dfrac{16}{2}(5 + 95) = 8(100)$.

Conclusion: $S_{16} = 800$.

Example 6 — Application — rising salary

A worker's salary starts at \$38,000 and rises by \$1400 each year. Find the total earned over the first 9 years.

Solution. $u_1 = 38000$, $d = 1400$, $n = 9$. $S_9 = \dfrac{9}{2}(2(38000) + 8(1400)) = \dfrac{9}{2}(76000 + 11200)$.

Conclusion: $S_9 = \dfrac{9}{2}(87200) = \$392{,}400$.

Example 7 — Solve for n from a sum equation

An arithmetic sequence has $u_1 = 2$, $d = 6$. Find n such that $S_n = 184$.

Solution. **Step 1:** $S_n = \dfrac{n}{2}(4 + 6(n-1)) = n(3n-1) = 3n^2 - n$.

Step 2: $3n^2 - n - 184 = 0 \Rightarrow (n-8)(3n+23) = 0$.

Conclusion: $n = 8$ (rejecting the negative root).

Example 8 — Two sequences meeting

Sequence A: $u_1 = 10$, $d = 3$. Sequence B: $u_1 = 50$, $d = -2$. Find the term number where both sequences give the same value.

Solution. **Step 1:** Set general terms equal: $10 + 3(n-1) = 50 - 2(n-1)$.

Step 2: $3n + 7 = 52 - 2n \Rightarrow 5n = 45 \Rightarrow n = 9$.

Conclusion: at $n = 9$, both sequences equal 34.

Example 9 — Application — accumulated temperature drop

A region's daily low starts at 20řC on day 1 and drops by 1.5řC each day. Find the sum of all daily lows over the first 14 days.

Solution. $u_1 = 20$, $d = -1.5$, $n = 14$. $S_{14} = \dfrac{14}{2}(2(20) + 13(-1.5)) = 7(40 - 19.5)$.

Conclusion: $S_{14} = 7(20.5) = 143.5$řC (sum of daily lows).

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find u_{15} for $6, 10, 14, \ldots$

Question 2

Find u_{30} for $50, 45, 40, \ldots$

Question 3

An arithmetic sequence has $u_4 = 18$ and $u_9 = 38$. Find u_1 and d.

Question 4

Find S_{18} for $2, 9, 16, \ldots$

Question 5

Find S_{40} for $100, 95, 90, \ldots$

Question 6

A theatre has 18 seats in the first row, increasing by 3 seats per row for 12 rows. Find the total seating capacity.

Question 7

Find the number of terms in the sequence $5, 9, 13, \ldots, 161$.

Question 8

An arithmetic sequence has $u_1 = 3$, $d = 5$, and last term $u_n = 98$. Find S_n.

Question 9

A stadium's rows increase by 4 seats each row, starting at 30 in row 1, for 20 rows. Find the total capacity.

Question 10 — Challenge

How many terms of $3, 7, 11, \ldots$ are needed for the sum to first exceed 200?

Question 11 — Challenge

An arithmetic sequence has $u_1 = 4$ and $S_{10} = 205$. Find d.

Question 12 — Challenge

An arithmetic sequence has $u_1 = 5$, $d = 8$. Find n such that $S_n = 612$.

Question 13 — Challenge

Sequence C: $u_1 = 100$, $d = -7$. Sequence D: $u_1 = 4$, $d = 9$. Find the term number and value where they meet.

Question Recap & Answer Key

Question Recap

1. Find u_{15} for $6, 10, 14, \ldots$
2. Find u_{30} for $50, 45, 40, \ldots$
3. An arithmetic sequence has $u_4 = 18$ and $u_9 = 38$. Find u_1 and d.
4. Find S_{18} for $2, 9, 16, \ldots$
5. Find S_{40} for $100, 95, 90, \ldots$
6. A theatre has 18 seats in the first row, increasing by 3 seats per row for 12 rows. Find the total seating capacity.
7. Find the number of terms in the sequence $5, 9, 13, \ldots, 161$.
8. An arithmetic sequence has $u_1 = 3$, $d = 5$, and last term $u_n = 98$. Find S_n.
9. A stadium's rows increase by 4 seats each row, starting at 30 in row 1, for 20 rows. Find the total capacity.
10. How many terms of $3, 7, 11, \ldots$ are needed for the sum to first exceed 200?
11. An arithmetic sequence has $u_1 = 4$ and $S_{10} = 205$. Find d.
12. An arithmetic sequence has $u_1 = 5$, $d = 8$. Find n such that $S_n = 612$.
13. Sequence C: $u_1 = 100$, $d = -7$. Sequence D: $u_1 = 4$, $d = 9$. Find the term number and value where they meet.

✓ Answer Key

1. $u_{15} = 62$
2. $u_{30} = -95$
3. $d = 4$, $u_1 = 6$
4. $S_{18} = 1107$
5. $S_{40} = -3700$
6. 12 rows: $S_{12} = \frac{12}{2}(2(18)+11(3)) = 6(69) = 414$ seats
7. $5 + (n-1)4 = 161 \Rightarrow n = 40$ terms
8. n: $98 = 3 + 5(n-1) \Rightarrow n = 20$; $S_{20} = \frac{20}{2}(3 + 98) = 10(101) = 1010$
9. $S_{20} = \frac{20}{2}(60 + 19(4)) = 10(136) = 1360$ seats
10. $S_n = \frac{n}{2}(6 + 4(n-1)) > 200 \Rightarrow 2n^2 + 4n - 200 > 0 \Rightarrow n^2 + 2n - 100 > 0$; testing, $n = 10$ gives $S_{10} = 210 > 200$ (and $n = 9$ gives $171 < 200$), so $n = 10$
11. $S_{10} = \frac{10}{2}(8 + 9d) = 205 \Rightarrow 40 + 45d = 205 \Rightarrow d = \frac{165}{45} = \frac{11}{3}$
12. $S_n = n(4n + 3) - n$... precisely $4n^2 + 3n = 612 \Rightarrow (n - 12)(4n + 51) = 0 \Rightarrow n = 12$
13. $100 - 7(n-1) = 4 + 9(n-1) \Rightarrow 107 - 7n = 9n - 5 \Rightarrow 112 = 16n \Rightarrow n = 7$; value $= 100 - 7(6) = 58$